

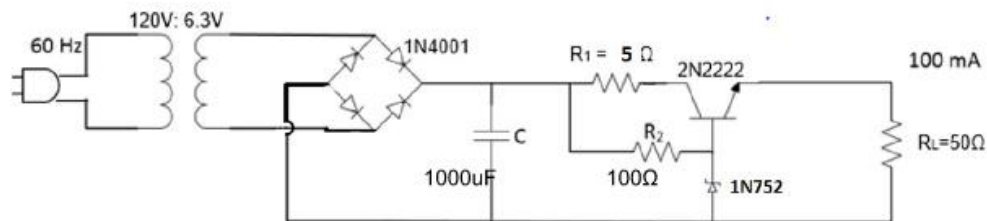
Experiment #13
AC-DC Power Supply Design Circuit

Executive Summary:

This experiment allowed us to take everything we learned from lecture and this lab to create an AC-DC power supply. Initially the circuit was built through simulation using PSPICE where we found a resistance for the base to emitter and the capacitance of the capacitor. Both elements were needed so that we could find the desired output voltage and signal. This output being 5VDC and less than 10mV of Vripple. Measuring each part of this circuit allowed us to see the behavior of the signal through the circuit and how by implementing what we have learned we can create our own power supplies.

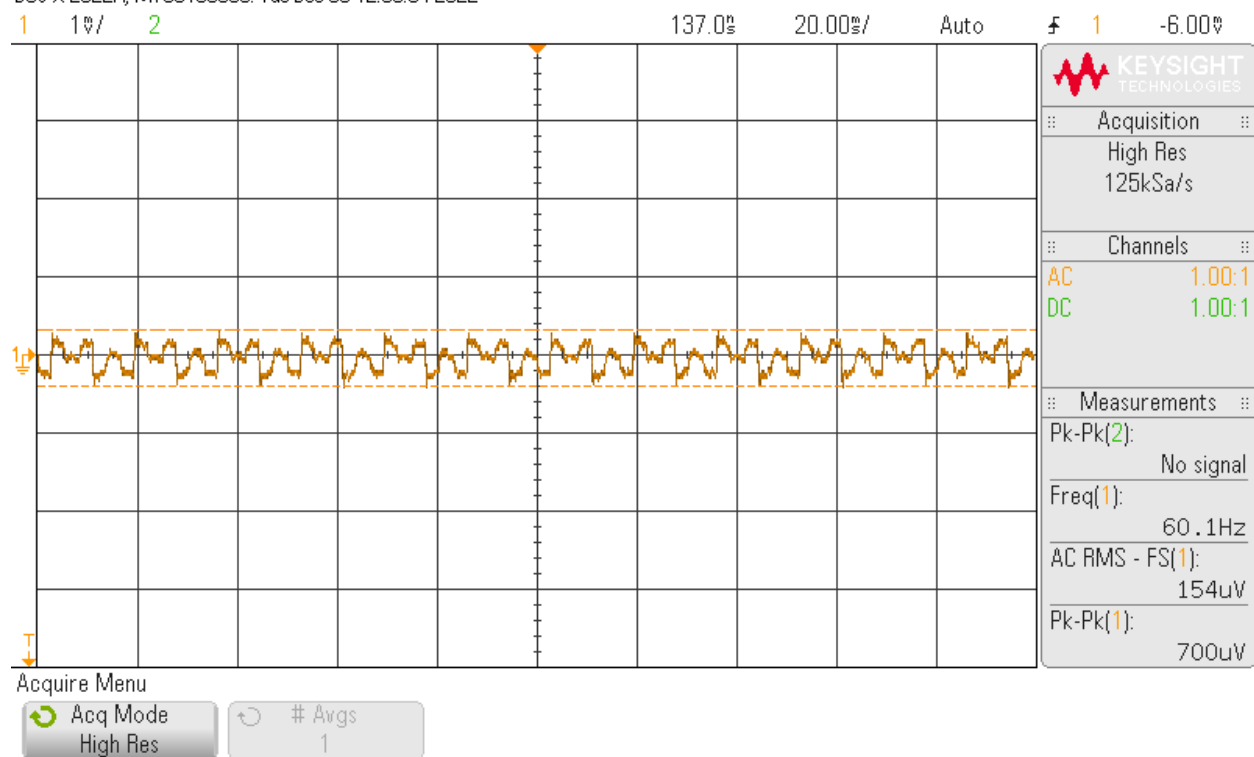
Objective: To design a simple AC-DC power supply using devices from experiments we used throughout the semester.

Diagrams:



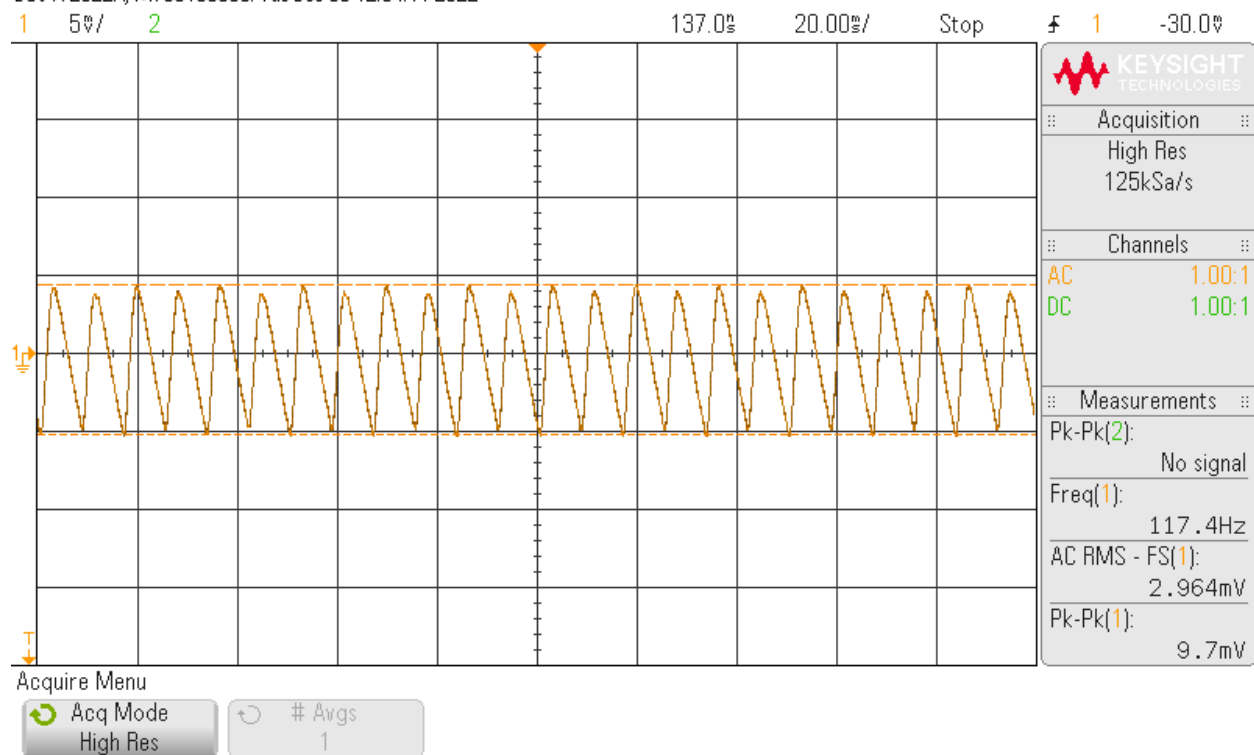
No load

DSO-X 2022A, MY53160365: Tue Dec 06 12:36:54 2022



Load

DSO-X 2022A, MY53160365: Tue Dec 06 12:34:11 2022



Analysis:

From the prelab, we built an AC-DC power supply through P-SPICE. The goal was to get an output Vripple across the load resistor. To find this we had to find a correct capacitor value as well as a resistor value. Our discovered values were 1000uF capacitor and a 100Ω resistor. These values gave us an output Vripple voltage of about 9.7mV which is below the 10mV across the load resistor.

Building the circuit in lab, we scoped from left to right first from the transformer, then the Wheatstone bridge, followed by capacitor and then a BJT. Then finally at the load resistor. The scope let us follow along and see what we have learned in class. 120VAC passed the transformer into the circuit. The windings create a ratio so in the scope what we see is stepped down 18V peak to peak.

Passing the Wheatstone bridge, the signal becomes fully rectified and the wave we see is positive at about a 9.4V peak to peak. After the bridge the capacitor shows us ripples which have been explored in other labs. These ripples are caused by the time delay from charging and discharging the capacitor. These ripples measured to be about 7.5V peak to peak. R2 was then used to turn on the BJT since it is a current controlled current device. Finally, the Zener diode smooths out our DC wave for us by getting rid of the ripple. The output voltage becomes 5V and our Vripple is less than 10mV. To turn on the BJT the base emitter voltage required was .7V and the Zener diode taking 5.6V is how we get that output. Whatever is left over is basically dumped on the load resistor. We can see the differences caused by load and no load above.